MATHEMATICS

Applications and interpretation

Higher Level | Set B | Paper 3

Challenge mock examination

Time allowed: 1 hour 15 minutes Maximum mark: 55

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• This paper contains two compulsory extended problem-solving investigations.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

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Question 1 | Question paper

Question 1 [Maximum mark: 27]

A laboratory heating device receives two current components with the same period of 12 seconds. Their combined current, in amperes, is modelled by

$$I(t) = 4\cos(\omega t) + 3\cos(\omega t - \phi), \qquad \omega = \tfrac{\pi}{6}, \qquad 0 \le \phi \le \pi.$$

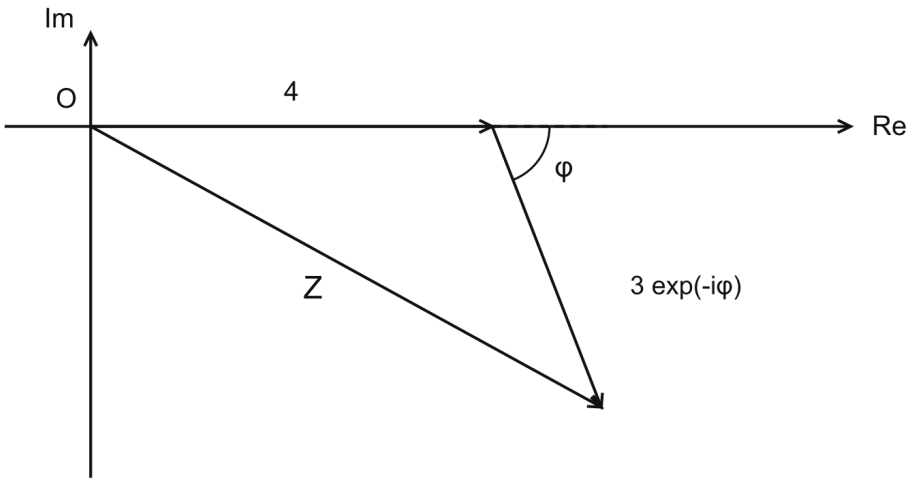
The adjustable phase is ϕ. Write the combined current as $I(t) = A\cos(\omega t - \delta)$ with $A > 0$. A phasor diagram is shown for one phase choice.

When current is supplied to the heater, its useful instantaneous heating power is $10I(t)^2$ watts. A safety controller diverts the current away from the heater whenever $|I(t)| > 4$ A. While diverted, the useful power is zero; the input current model itself is unchanged. Let q be the fraction of a complete period for which the heater is off.

For any periodic power function $p(t)$ with period 12 s, its mean power is

$$\overline{P} = \tfrac{1}{12}\int_0^{12} p(t)\,\mathrm{d}t.$$

You may use $\cos^2 u = (1 + \cos 2u)/2$. All angles are in radians.



Phasor construction for one phase choice; all phases in radians.

(a) For $\phi = 1.2$, use the complex number $Z = 4 + 3e^{-i\phi}$ to find A and δ, giving $0 \le \delta < \pi$. [3]

(b) For variable ϕ, show that $A^2 = 25 + 24\cos\phi$. Hence state the possible range of A. [3]

Question 1 | Question paper

(c) If the safety controller were removed, show that the mean heating power over a complete period would be $5A^2$ watts. [3]

(d) For $A > 4$, obtain q in terms of A. Also state q when $1 \leq A \leq 4$. [4]

(e) The permitted fraction of time off is at most $1/4$. Find the resulting interval of permissible phases ϕ. [3]

(f) With the controller operating, obtain an expression for the mean useful power $P(A)$ for $1 \leq A \leq 4$, and for $4 < A \leq 7$. Show the integration leading to your expression. You may use symmetry over one complete phase cycle. [6]

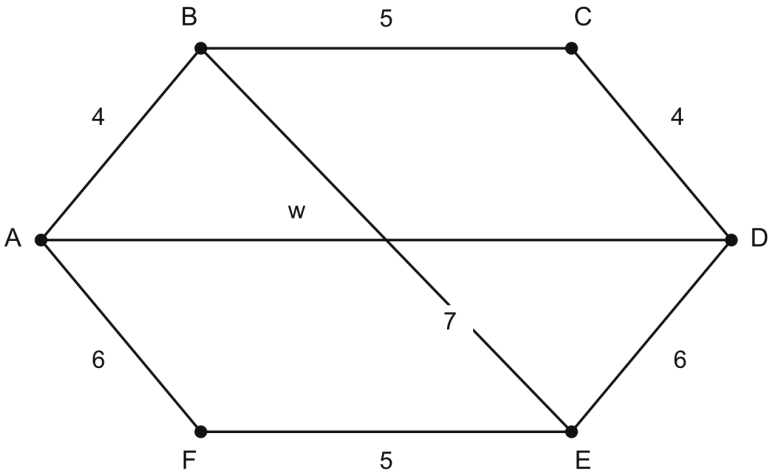
(g) Use technology to investigate $P(A)$ under the restriction in part (e). Find the phase that maximizes mean useful power and the resulting maximum. Support your conclusion over the whole permitted domain. Explain why choosing the largest permitted amplitude is not optimal. [5]

Question 2 | Question paper

Question 2 [Maximum mark: 28]

A maintenance crew must inspect every link of an underground service network. The undirected graph has the eight edges shown; edge weights are traversal times in minutes. Inspecting a link requires traversing the whole edge at least once. Edges may be repeated, and there are no junctions at crossings in the diagram.

The time on link AD is a positive parameter w. All other times are fixed. Unless a part states otherwise, the crew starts and finishes at A. The aim is to minimize total traversal time, including repeated edges. Denote this minimum closed-route time by $L_c(w)$.



Edge weights are minutes. Crossings of edges are not junctions.

(a) Identify the odd-degree vertices. Explain why the crew cannot inspect all eight links exactly once on a closed route. [2]

(b) For $w = 2$, construct a table of shortest traversal times between the odd-degree vertices. Show any indirect paths used. [3]

(c) For $w = 2$, find the minimum closed-route time and give one route attaining it. Justify why a shorter closed route is impossible. [5]

Question 2 | Question paper

(d) Investigate how the optimal repeated links change as w varies. Obtain an exact piecewise expression for $L_c(w)$ for all $w > 0$, justifying the breakpoint and all candidate pairings. [6]

(e) The link AD is removed from the inspection network. Find the new minimum closed-route time. Explain why removing a useful connection can nevertheless shorten an inspection route. [3]

(f) Instead, keep all eight links, but allow the crew to start at A and finish at D. Show that the minimum open-route time is $L_o(w) = 44 + w$ for all $w > 0$. [4]

(g) An open route incurs an additional return cost equivalent to $h > 0$ minutes. Derive a complete decision rule in terms of w and h for choosing a closed route or an open route plus return. Include all tie cases. Explain why the maximum possible saving in traversal time from using an open route is only 3 minutes, however large w becomes. [5]

End of question paper

MATHEMATICS

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Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

Question 1 [Maximum mark: 27]

(a) [3 marks]

A1 $Z = 5.087073263\ldots - 2.796117258\ldots i$.

A1 $A = |Z| = 5.804876063\ldots \approx 5.80$ A.

A1 $\delta = -\arg Z = 0.5025755802\ldots \approx 0.503$ radians, so the combined cosine is delayed by δ.

(b) [3 marks]

M1 $|Z|^2 = (4 + 3\cos\phi)^2 + (-3\sin\phi)^2$.

A1 $A^2 = 16 + 24\cos\phi + 9(\cos^2\phi + \sin^2\phi) = 25 + 24\cos\phi$.

A1 As ϕ ranges from 0 to π, $1 \le A \le 7$.

(c) [3 marks]

M1 Use the average $\frac{10A^2}{12}\int_0^{12}\cos^2(\omega t - \delta)\,\mathrm{d}t$.

M1 Using the provided identity, the oscillatory cosine term has integral zero over a whole period and the constant term integrates to 6.

A1 The unrestricted mean power is $5A^2$ W, independent of δ.

(d) [4 marks]

M1 For $c = 4/A \in (0, 1)$, the controller is off when $|\cos u| > c$.

A1 Over a phase interval of length 2π, this has total length $4\arccos c$.

A1 $q = 2\arccos(4/A)/\pi$ for $A > 4$.

A1 For $1 \le A \le 4$, the threshold is never exceeded and $q = 0$.

(e) [3 marks]

M1 $2\arccos(4/A)/\pi \le 1/4$ gives $A \le 4/\cos(\pi/8)$, including the $A \le 4$ cases.

A1 $A \le 4.329568801\ldots$; use $25 + 24\cos\phi \le 16\sec^2(\pi/8)$.

A1 $\arccos((16\sec^2(\pi/8) - 25)/24) \le \phi \le \pi$, or $1.834458682\ldots \le \phi \le \pi$.

Question 1 | Worked solutions

(f) [6 marks]

M1 A full-period average is unaffected by the phase shift; work with u over $0 \le u \le 2\pi$.

M1 For $A > 4$, put $c = 4/A$ and use symmetry: $P = \frac{10A^2}{2\pi}\, 4 \int_{\arccos c}^{\pi/2} \cos^2 u \, \mathrm{d}u$.

M1 An antiderivative is $u/2 + \sin(2u)/4$.

A1 The integral of $\cos^2 u$ over all on-intervals is $2 \arcsin c - 2c\sqrt{1-c^2}$.

A1 $P(A) = \frac{10A^2}{\pi}\left[\arcsin(4/A) - \frac{4}{A}\sqrt{1 - \frac{16}{A^2}}\right]$ W for $4 < A \le 7$.

A1 For $1 \le A \le 4$, there is no shutdown and $P(A) = 5A^2$ W. Both expressions approach 80 W at $A = 4$.

(g) [5 marks]

M1 Investigate the piecewise function on the full permitted interval $1 \le A \le 4\sec(\pi/8) = 4.329568801\ldots$.

A1 The maximum is 80 W at $A = 4$.

M1 Support globality with a labelled graph or equivalent numerical/analytic investigation showing increase up to 4 and decrease from just above 4 to the upper bound. A single isolated calculator maximum is insufficient.

A1 $\phi = \arccos((16-25)/24) = \arccos(-3/8) = 1.955193101\ldots \approx 1.96$ radians.

R1 Above $A = 4$, the largest instantaneous powers are discarded by shutdown. Increasing amplitude increases lost peak intervals; a larger permitted amplitude therefore produces less useful mean power.

Question 2 | Worked solutions

Question 2 [Maximum mark: 28]

(a) [2 marks]

A1 A, B, D, E each have degree 3; C, F have degree 2.

R1 A closed Eulerian route requires every degree even. Four odd vertices prevent traversing every edge exactly once on a closed route.

(b) [3 marks]

M1 Use shortest paths, including $A \to D \to E$ for AE and $B \to A \to D$ for BD.

A2 For the order (A, B, D, E), shortest distances are $d_{AB} = 4$, $d_{AD} = 2$, $d_{AE} = 8$, $d_{BD} = 6$, $d_{BE} = 7$, $d_{DE} = 6$; diagonal zeros and symmetric entries complete the table.

(c) [5 marks]

M1 Compare the three odd-vertex pairings: (AB, DE), (AD, BE) and (AE, BD).

A1 Their added costs are 10, 9 and 14, so repeat AD and BE.

M1 The original edge total is $37 + w = 39$; any closed inspection route must correct all four odd degrees, requiring at least the least pairing cost.

A1 A route attaining it is $A - B - E - F - A - D - C - B - E - D - A$.

A1 The minimum is $39 + 9 = 48$ minutes.

(d) [6 marks]

M1 For general w, $d_{AD} = \min(w, 13)$, since the alternative shortest path is $A - B - C - D$.

A1 The pairing (AB, DE) always costs $4 + 6 = 10$.

A1 The pairing (AD, BE) costs $\min(w, 13) + 7$.

A1 The third pairing costs $\min(w + 6, 11) + \min(w + 4, 9)$, which is $10 + 2w$ for $0 < w \leq 5$ and 20 for $w \geq 5$.

R1 The third pairing is never cheaper than 10. The first two tie at $w = 3$; repeat AD, BE below 3, or AB, DE above 3.

A1 $$L_c(w) = 37 + w + \min(w + 7, 10) = \begin{cases} 44 + 2w, & 0 < w \leq 3, \\ 47 + w, & w \geq 3. \end{cases}$$

Both formulas give 50 at $w = 3$.

(e) [3 marks]

M1 After removing AD, only B, E are odd and the remaining edge total is 37.

A1 Repeat shortest path BE of length 7, giving $37 + 7 = 44$ minutes.

R1 The inspection obligation itself loses an edge, and the parity requirements change. A connection may shorten point-to-point travel while increasing the work required to inspect every link.

(f) [4 marks]

M1 An open route from A to D requires exactly A, D to remain odd.

A1 Thus the extra traversals need only correct the odd degrees at B, E.

A1 The shortest B to E path is the direct edge of length 7 for every $w > 0$.

A1 $L_o(w) = 37 + w + 7 = 44 + w$. For example $A - B - E - F - A - D - C - B - E - D$ attains this total.

(g) [5 marks]

M1 $L_c(w) - L_o(w) = \min(w, 3)$.

A1 The open option including return costs $44 + w + h$; its saving relative to the closed option is $\min(w, 3) - h$.

R1 Choose open strictly when $0 < h < 3$ and $w > h$. Choose closed strictly when $h > \min(w, 3)$.

A1 Tie when $h = \min(w, 3)$: equivalently $w = h \leq 3$, or $h = 3$ and $w \geq 3$.

R1 For $w \geq 3$, the best closed route repeats AB and DE for 10 minutes, while the open route repeats BE for 7. This fixed 3-minute difference bounds the saving; increasing w adds the same once-traversed link time to both routes.

End of worked solution book